



EUROPEAN MICROSOFT FABRIC S&L Community Conference

BARCELONA
- 01 OCT 2026



From Notebook to Bedside: Building a Clinical Risk Prediction Pipeline in Fabric



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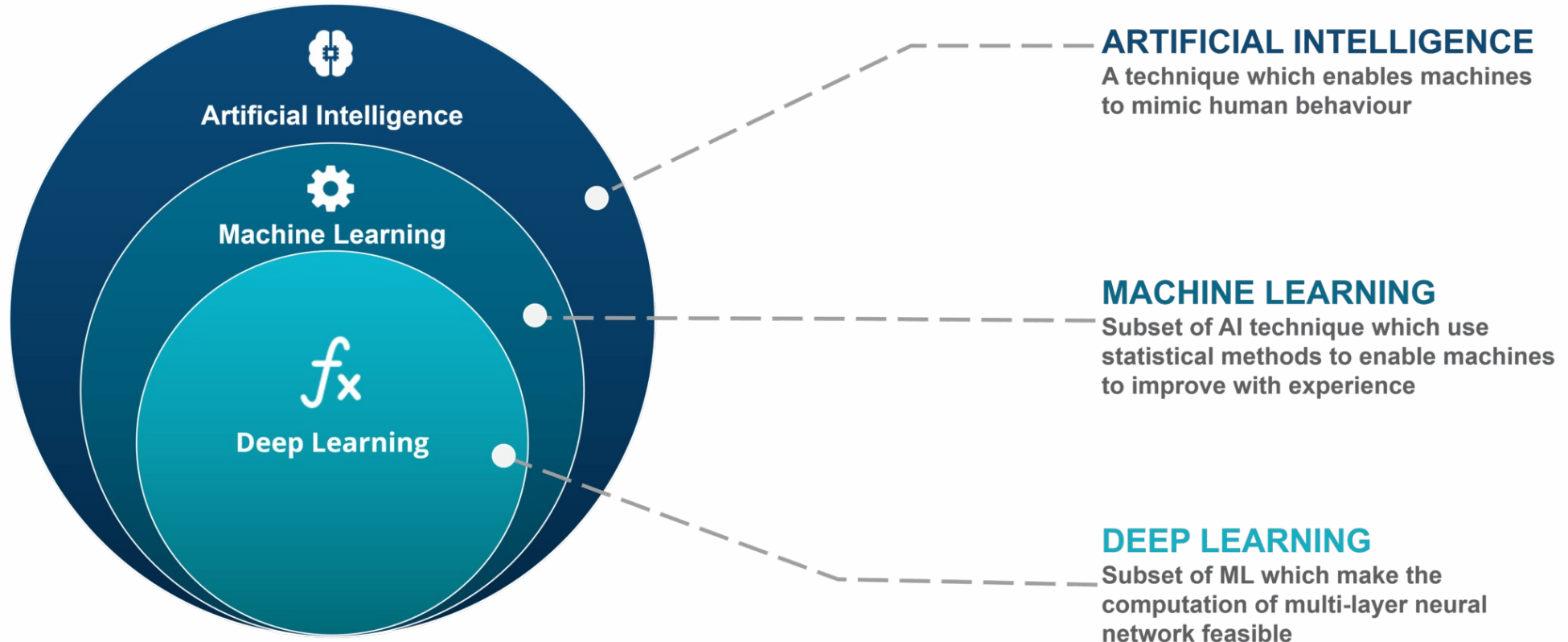
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Agenda

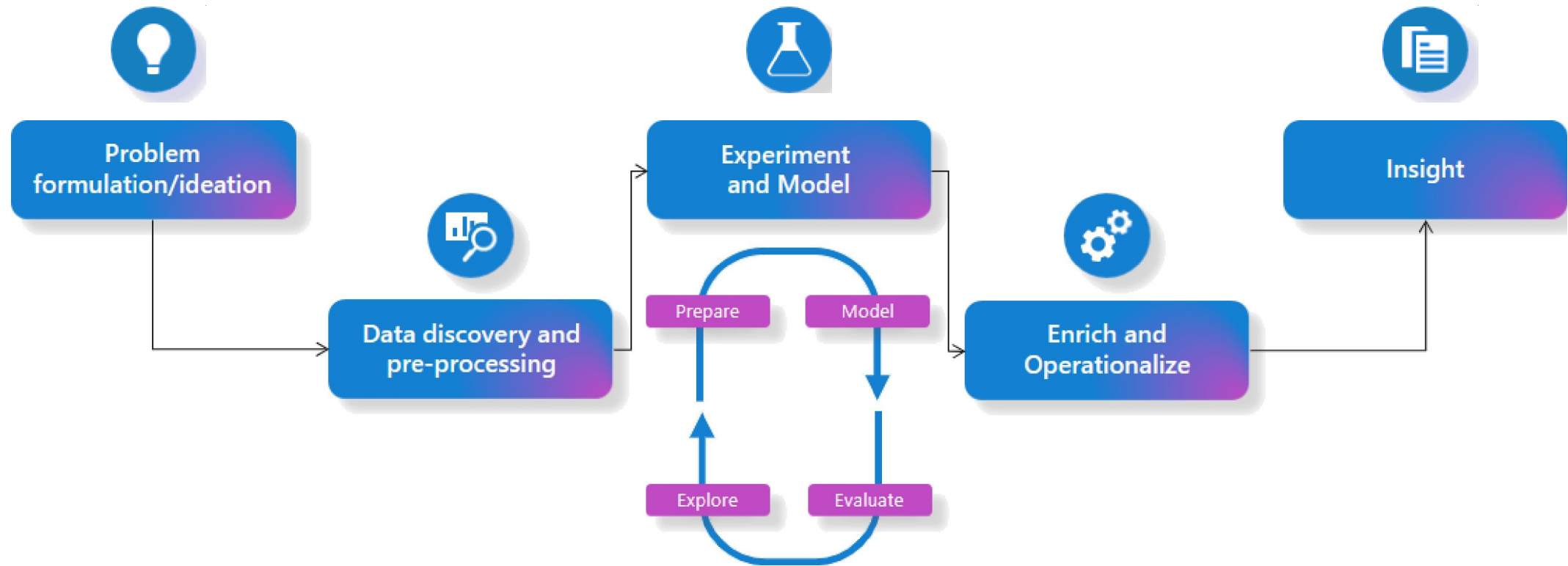
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| 04 | Governed ML | XX |
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Fabric Data Science & Machine Learning

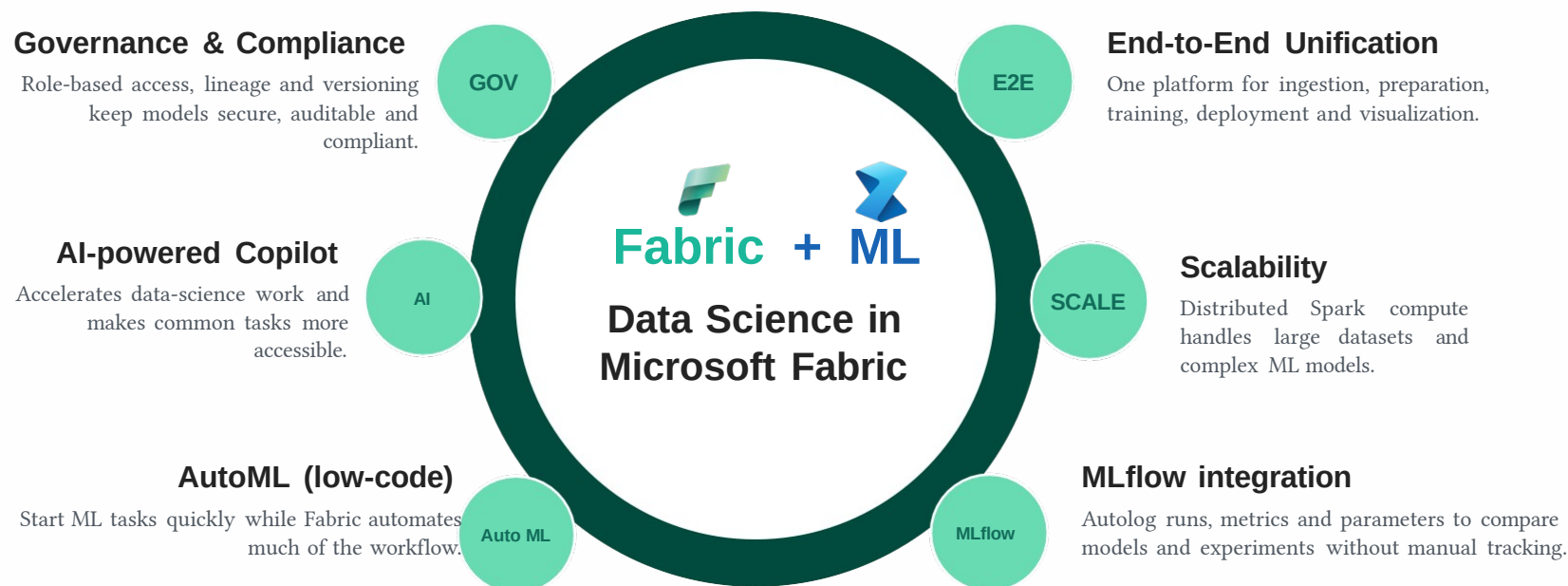
ML as a subset of AI



The data science process



Microsoft Fabric as a platform for Data Science

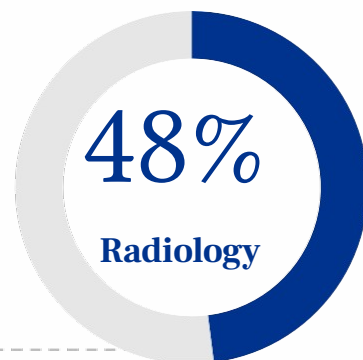


Prediction Models in Healthcare

AI and ML is being embedded in EU healthcare systems

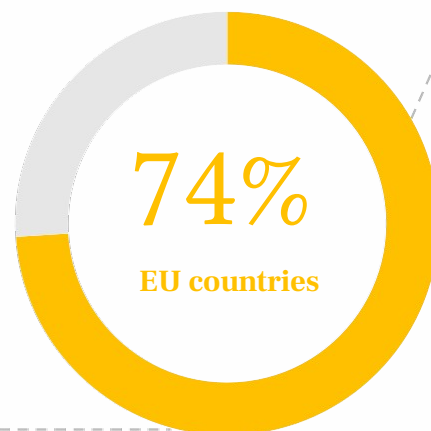
Radiology reality

48% of 572 European radiology respondents used AI; another 25% planned adoption.



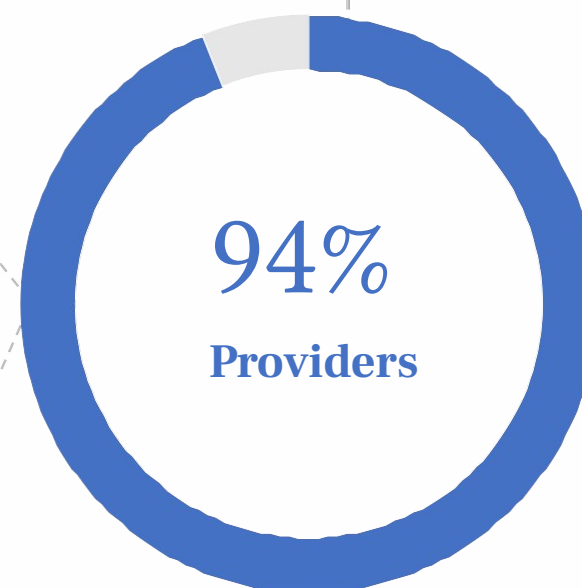
Diagnostics spread

Nearly 3 in 4 EU countries report AI-assisted diagnostics in clinical settings.



Provider pipeline

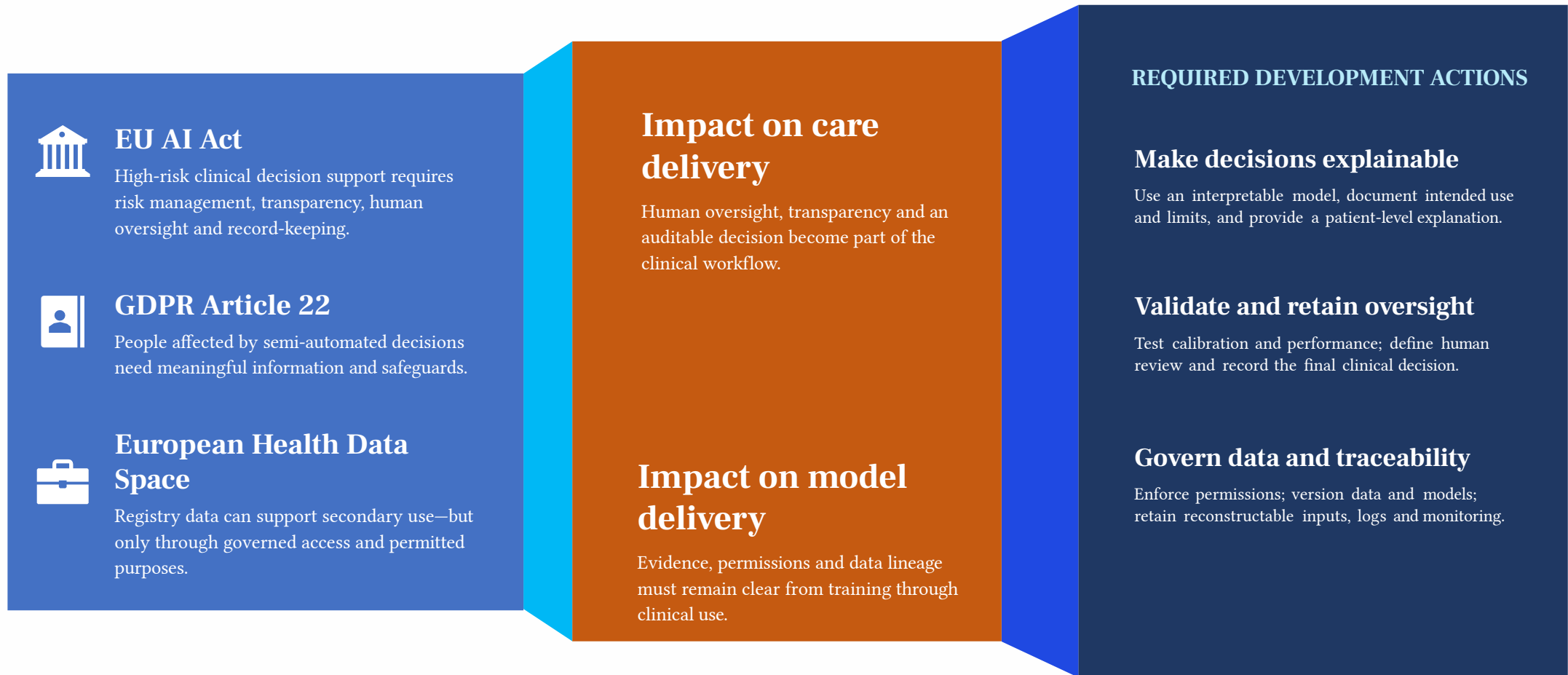
94% of surveyed EU healthcare providers already use AI solutions or plan to adopt them.



Measurement caveat: no harmonized EU hospital-level adoption ranking exists; these indicators use different denominators and are not directly comparable.

Sources: WHO/Europe EU readiness survey (2024–25); European Commission Digital Health Observatory (2026); Zanardo et al., Insights into Imaging (2024).

However, EU rules turn healthcare ML into a governed clinical process



Yet not all institutions are prepared...

Box E3. Legal and regulatory landscape for AI in health: key findings

7%

(2 of 27) have issued guidance for manufacturers and users on applying existing liability regimes to AI



48%

(13 of 27) have implemented data accountability practices as a minimum standard for AI governance in health



56%

(15 of 27) have plans to align legal liability standards with upcoming EU legislation



22%

(6 of 27) have issued practical guidance on ethics by design for AI systems in health



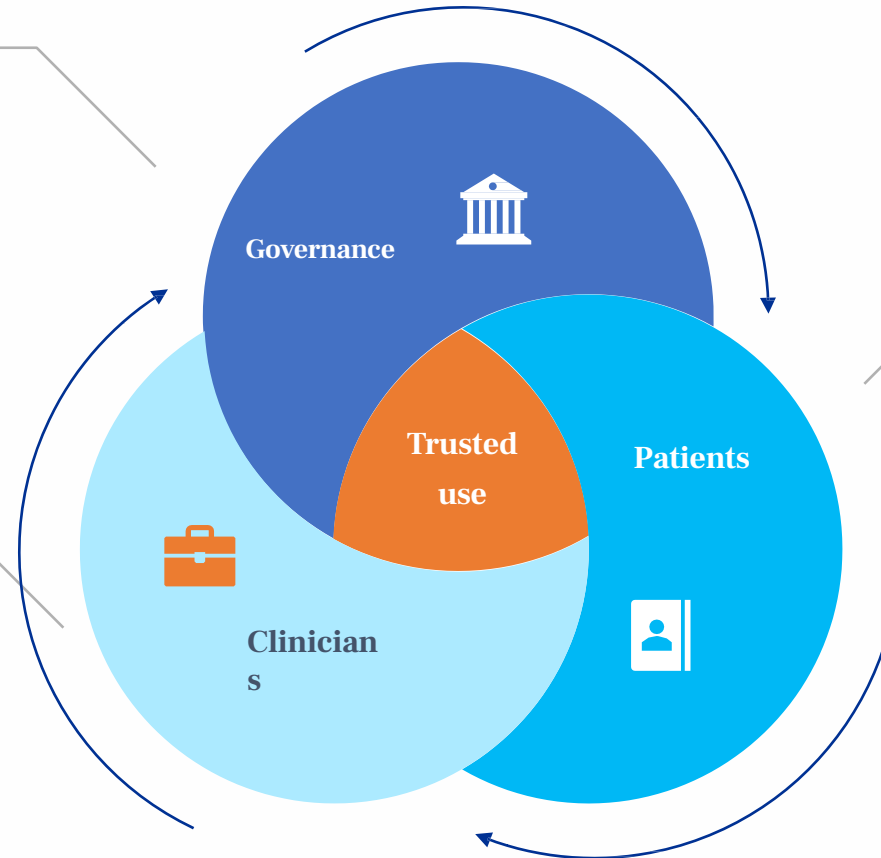
Three audiences face different obligations

Regulators & governance

Need auditable lineage, versioning, monitoring and a recorded human decision for every use.

Clinicians

Need an interpretable, calibrated score in an approved workflow—while retaining final accountability.



Patients

Need a clear explanation of the score, its uncertainty and how it influenced—or can be challenged in—care.

Fabric Healthcare Data Solutions

Healthcare data solutions

Improve the efficiency and value of your healthcare data

These healthcare data solutions help you unify disparate data sources running on Fabric. Each capability contains a unique set of tools to help you prepare the data for ingestion, transformation and harmonization for advanced analytics and AI modeling.

Deploy the relevant solutions to your workspace to get started on your healthcare data transformation goals.



Setup your solution to get started

The setup module will set up your healthcare data estate. You can select, configure and deploy capabilities such as data connectors and sample data in a guided, step-by-step process.

[Setup your solution](#)

Solution capabilities and sample data

Select each capability to learn about it and build your data transformation solution.



Sample data

Test your healthcare data solutions using this sample data.



Azure health data services - data export

Export data from the FHIR service in Azure Health Data Services to OneLake in Fabric.



CMS Claims data transformations

Bring your CMS CCLF (Claim and Claim linefeed) data to OneLake.



DICOM data transformation

Bring your DICOM (Digital Imaging and Communications in Medicine) imaging data to OneLake.



OMOP transformations

Prepare data for standardized analytics through OMOP (Observational Medical Outcomes Partnership) open community standards.



SDOH datasets - transformations

Bring social determinants of health (SDOH) datasets to the OneLake to unify it with core healthcare data domains such as clinical and claims.



Dynamics 365 Customer Insights - data preparation

Connect Dynamics 365 Customer Insights to your OneLake on Fabric for creating patient or member lists for your outreach.



Patient outreach analytics

Bring marketing and analytics data from Dynamics 365 to OneLake. Harmonize this data with your patient data and leverage our Power BI templates to improve patient engagement.



Care management analytics

An analytical template that combines data from various key modalities of healthcare data solutions to provide insights and make data driven decisions.



DAX Copilot data transformation

Prepare DAX Copilot data for analytics



Orchestrate multimodal AI insights (Public Preview)

Enable on-demand generation and integration of AI enrichments from our 1P healthcare AI models and APIs.

The case: knee replacement on NHS PROMs

NHS PROMs: what is actually in it

- THE DATA

- NHS PROMs national data pack, knee replacement cohort, anonymised at source
- One row per patient episode; roughly 46,000 procedures a year across the collection window
- Pre-operative questionnaire: Oxford Knee Score 0–48 with pain, function and ADL subscales; EQ-5D-3L; symptom period; previous surgery; living arrangements; mobility
- Twelve comorbidity flags
- Provider dimension joined in: ODS code, university / independent status
- Post-operative OKS at six months — when the patient returns the questionnaire

- THE PREDICTION

- Inputs are pre-operative only — everything knowable before the decision to operate
- Target is derived: $\text{oks_delta} = \text{oks_t1} - \text{oks_t0}$
- The positive class is the BAD outcome — the patient who does not meaningfully improve, because that is the class worth detecting
- ~45 features after encoding
- Anonymised, national, already public-sector governed. Every interesting problem is downstream of that.
- [TODO: settle the target definition and prevalence — see notes]

Charts

Our Support

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Process

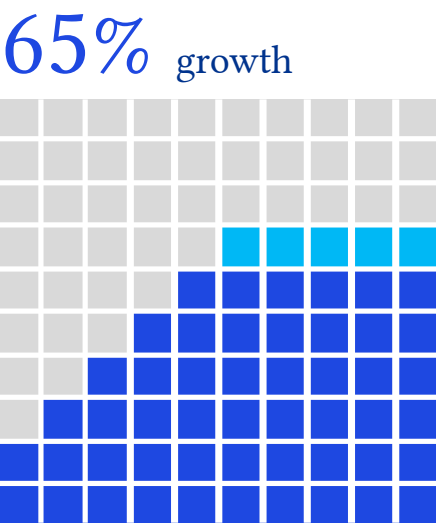
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Target

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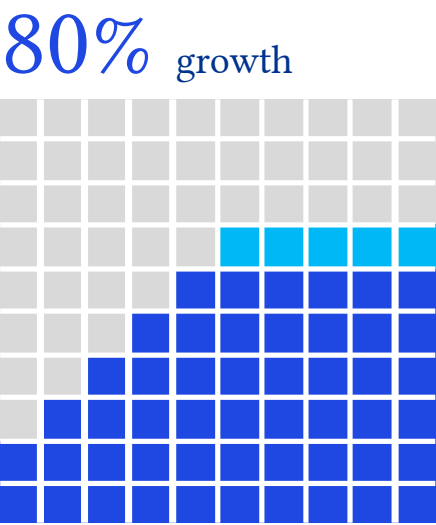


Subtitle

Dolor sit amet, consectetur adipiscing elit sed pulvinar tortoramet vulputate tellus.

Lorem ipsum

Dolor sit amet, consectetur adipiscing elit. Praesent sed pulvinar tortor, sit amet vulputate tellus.



Subtitle

Dolor sit amet, consectetur adipiscing elit sed pulvinar tortoramet vulputate tellus.

Why 7 points?

- The Oxford Knee Score runs 0 – 48. A 2-point rise is real arithmetic and no change the patient can feel.
- The minimal clinically important difference — the smallest change a patient notices — is around 7 points.
- [CHART V7: two panels, exported from 31_data_exploration §2.
- Left: histogram of oks_delta, 60 bins, MCID line drawn at 7, region below it shaded.
- Right: mean OKS gain against pre-operative score — it falls as the starting score rises.]



The data problems that shaped the pipeline

- IMBALANCE
- [CHART V8: class balance bar from 31_data_exploration §4.
- Print the target definition next to the prevalence.]
- → Precision–recall, never accuracy. A model that says good outcome to everyone
- TWO MISSINGNESS MECHANISMS
- [CHART V9: missingness bars in two labelled groups with a visible gap.
- Group A, suppressed by NHS disclosure control: age_band, gender — missing on exactly the same rows



Let's look at the data

- 1. The 7-point rule and the delta distribution
 - 2. Knee vs hip — why this cohort
 - 3. Class balance
 - 4. The patients with no outcome at all
 - 5. The sentinel trap
-
- Show in Fabric: notebook 31_data_exploration, pre-run, in the 35_Exploration folder — reading silver.knee_episode and gold.knee_features directly. Nothing in it writes.

Setting up train and test without leaking

- THE ORDER MATTERS

- Stratified split before any imputation or encoding
- MICE (miceforest) builds a model per column with missing values — fit it on everything and test-set information flows into your training features
- MICE kernel + one-hot encoder wrapped in one

- THE LEAKAGE GUARD

- [FILL: paste the two assertions from 40_train_register §3 — 4–5 lines, monospace]
- Two assertions before a single model is fitted. A reviewer would catch both eventually; catching them here means they never reach



A model a clinician can argue with

What we trained, and what it actually does

- Held-out test set, threshold 0.8:
- Logistic regression — 18 flagged, precision 1.000, recall 0.003
- Random forest — 158 flagged, precision 0.969, recall 0.033, 5 FP
- EBM (shipped) — 169 flagged, precision 0.941,

- [TABLE V7: build as a native PowerPoint table with tabular figures, EBM row tinted.
- Columns: model / flagged / precision / recall / F1 / false positives / interpretability.
- Interpretability values: glassbox, post-hoc SHAP, glassbox.
- [TODO: add the PR-AUC / average-precision column once computed]]



Global, local, and the contrast

- Global — the shape function for pre-operative OKS, and one comorbidity
 - Local — six ranked contributions for one pinned pre-operative patient
 - Contrast — the same patient through the random forest, via SHAP
-
- $\text{risk} = \text{sigmoid}(\text{intercept} + \sum f(x))$. The bars on screen are those terms — no surrogate, no approximation.



Calibration, and where we put the line

- A NUMBER THAT MEANS WHAT IT SAYS
- [CHART V10: reliability curve for the calibrated EBM — decile bins, diagonal reference, bin counts annotated.]
- Of a hundred patients scored 34%, about thirty-four should
- AND WHERE WE PUT THE LINE
- [CHART V11: threshold sweep from 42_threshold_analysis — precision and flagged count against threshold, 0.5 → 0.8, marker on 0.8.]
- [FILL: recompute the weekly arithmetic once the prevalence



Tracking that survives an audit

- THE QUESTION, TWO YEARS LATER
- “Reconstruct the exact input this model was trained on, and show me who approved it.”
- Most teams cannot — not from carelessness, but because the answer was never written anywhere
- THE TAGS THAT MATTER
- gold_table + gold_table_version — the Delta version. “Trained on gold” becomes “gold as of version 7”, which time travel reconstructs exactly. This is the one they ask for.
- feature_set_version — bump when the feature list



Live: the experiment and the registered model

- 1. The ML experiment item — one parent run per training session, one child run per model. Collapse to compare sessions, expand to compare models.
- 2. Filter the run list by tag: interpretability = blackbox. That is the deployment gate, as a query.
- 3. The registered ML model item — versions, and the tags that travelled with each one.
- 4. gold_table_version on the shipped run → the Delta version it was trained on.
- 5. `SELECT * FROM gold.knee_features VERSION AS OF <n>`
— the training input, reconstructed live.

Delivery: endpoint, clinician app, agent

What the endpoint returns

- episode_id
- risk — calibrated
- threshold, flagged
- contributions[6] — feature and signed value
- model_version
- gold_table_version
- Versioned, authenticated, in-
- [FILL: paste the actual response payload as monospace code.
- Highlight model_version and gold_table_version in the accent colour.]
- [SCREENSHOT: the live call and its response — as a backup still.]
- Show in Fabric: the managed real-time endpoint on the registered model, and the

The clinician's end of the pipeline

- Seven decorated TypeScript classes become: a SQL database in Fabric, a GraphQL API through Data API Builder, Entra ID sign-in, and static hosting — all inside the workspace.

- `npx rayfin up --workspace "Healthcare-FabCon2026-Demo"`

- [DIAGRAM V13, left half: one decorated entity class from rayfin/data/ — ten lines of real code, unedited.]
- [DIAGRAM V13, right half: the four provisioned objects, drawn inside the workspace boundary from the 'three audiences' slide.]
- [SCREENSHOT: the deployed app — inbox, then patient detail with the contribution

Row-level security, demonstrated rather than claimed

- THE POLICY
 - policy: (claims, item) => claims.sub.eq(item.assigned ClinicianId)
 - Patient-scoped entities carry a policy: PatientRisk, RiskExplanation, ReviewDecision.
- WHERE IT IS ENFORCED
 - [DIAGRAM V15: two boxes, one struck through.
 - “Filter in the frontend” — struck through.
 - “Enforced in Data API Builder” — kept.]
 - RayfinDataService contains no WHERE clause on clinician.

- [DIAGRAM V16: grounding path in one row —
- question → Data Agent → Fabric IQ semantic layer → gold.patient_risk + gold.risk_explanation + silver.provider → cited answer.
- Reuse the workspace boundary from the 'three audiences' slide, and mark the permission check on the way IN, not on the way out.]

A grounded answer, or none

- “Which of my pre-operative patients are flagged, and why?”
- “Which factors contributed most for this patient?”
- “How does our flag rate compare with other providers?”

The missing layer, built

- 1,872 papers, 56 models in practice. The gap is not modelling capability.
- Data, analytics and decision in one governed place, with one lineage from a raw registry row to a recorded clinical decision.
- [DIAGRAM V6 AGAIN — same geometry as the 'three audiences' slide, now annotated.
- Clinician: the app, the six contributions, the recorded decision.
- Patient: calibrated probability, local explanation.
- Regulator: Delta version, sensitivity label, lineage, glassbox tag, RLS policy.

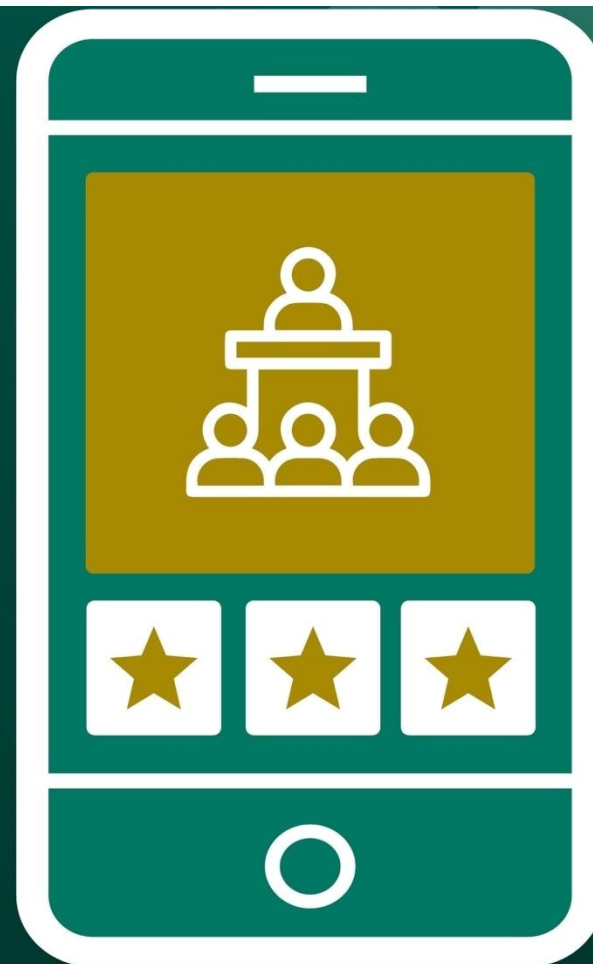


Five things that transfer

- Make every table Delta and log gold_table_version as a tag. One line of work, and your record-keeping obligation is met.
- Turn Fabric autologging off and log explicitly.
- Make interpretability a tag, so it can be a gate.
- [SCREENSHOT: QR code to the repo, large.]
- Ten Fabric notebooks, the medallion and MLflow design notes, the clinician app with its offline mode, and the model card.
- [FILL: both presenter handles]



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